

Joint Modeling of Longitudinal and Survival Data via Second-Order ODEs

A Mechanistic Approach to Biomarker Dynamics and Event Risk

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September 2025

[Time: 1 min — Total: 1 min]

Good afternoon, everyone. Thank you for joining me today.

I'd like to present our work on a new approach to joint modeling. We use second-order differential equations to model longitudinal biomarker data together with survival outcomes.

Our central question is this: Can we improve risk prediction by modeling not just biomarker levels, but also their velocities - how fast they're changing?

I'll walk you through our methodology, show some preliminary results, and would very much appreciate your feedback.

Motivating Example: PSA Velocity in Prostate Cancer¹

Goal: Joint modeling of biomarker trajectories and survival outcomes.

PSA Kinetics - Patient A

- Baseline PSA: 4.0 ng/mL
- PSA velocity: 0.35 ng/mL/year
- Pattern: Stable rise
- Low risk progression

PSA Kinetics - Patient B

- Baseline PSA: 4.0 ng/mL
- PSA velocity: 2.0 ng/mL/year
- Pattern: Rapid rise
- High risk progression

Can we leverage biomarker velocity to improve risk prediction?

¹D'Amico AV, et al. JAMA. 2005;294(4):440-447.

└ Motivating Example: PSA Velocity in Prostate Cancer¹

[Time: 2.5 min — Total: 3.5 min]

Let me start with a motivating example from clinical practice.

Consider two prostate cancer patients. Both have PSA levels of four point zero nanograms per milliliter today. Looking at just this snapshot, they seem identical.

But here's the key difference: Patient A's PSA rises at zero point three five units per year, while Patient B's increases at two point zero per year - nearly six times faster.

Clinically, these patients require very different management. Patient B needs more aggressive intervention.

This raises an important statistical question: How can we formally incorporate this velocity information into our models? Traditional joint models focus primarily on current levels. We hypothesized that explicitly modeling velocity could enhance predictive accuracy.

Goal: Joint modeling of biomarker trajectories and survival outcomes.

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Can we leverage biomarker velocity to improve risk prediction?

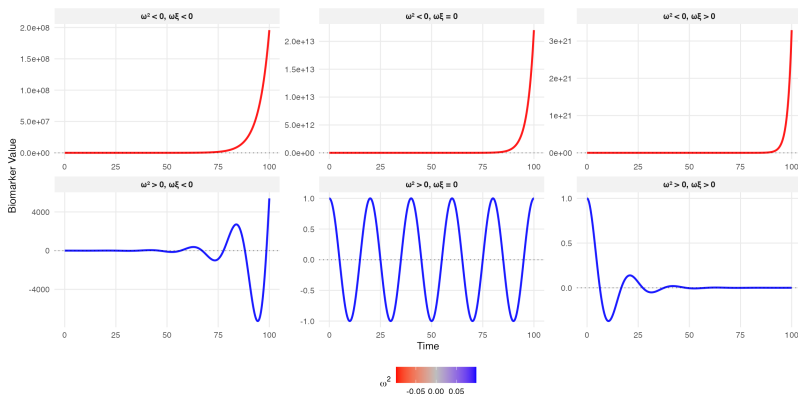
¹D'Amico *et al.* *JAMA*. 2005;294(4):446-457.

Our Solution: Mechanistic Modeling via ODEs

Second-order ODE framework for biomarker dynamics:

$$\ddot{m}(t) + 2\xi\omega\dot{m}(t) + \omega^2 m(t) = k\omega^2 u(t)$$

ξ : damping ratio, ω : natural frequency, k : gain, $u(t)$: excitation



JointODE

└ Motivation

└ Our Solution: Mechanistic Modeling via ODEs

[Time: 3.5 min — Total: 7 min]

We adopt second-order ODEs from engineering - and it turns out they're remarkably natural for modeling biological systems.

Look at the diversity of trajectories (tra-JECK-tor-eez) in this figure. These aren't just abstract mathematical curves - we observe exactly these patterns in real data: oscillatory behavior, smooth convergence to equilibrium, and divergent trajectories (tra-JECK-tor-eez).

Just three parameters capture the entire dynamics: - omega: The natural frequency - how quickly the system responds - xi (ksai): The damping ratio - whether we get oscillations or smooth decay - k u(t): The equilibrium level - where the biomarker is heading

Now here's what makes ODEs particularly powerful: velocity comes for free. Traditional methods require numerical differentiation. But when we solve the ODE, we automatically get both the biomarker level and its rate of change.

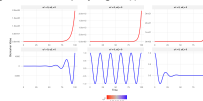
But there's more. We discovered that stability itself carries medical meaning. Stable trajectories (tra-JECK-tor-eez) indicate treatment is working, like this blue one here, while unstable ones signal trouble - perhaps resistance or progression, like the other curves.

[Click 2] The mathematics provides precise stability conditions: we need omega-squared greater than zero and omega-times-xi greater than zero.

But here's where it gets exciting: when these stability conditions fail, trajectories (tra-JECK-tor-eez) diverge. We can test for this instability in the data. However this is not today's focus. Maybe in future work. Today, I'll focus on the use of velocity in risk prediction.

$$m\ddot{y}(t) + 2\xi\omega_n\dot{y}(t) + \omega_n^2 y(t) = k u(t)$$

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Stability Conditions:

- **Stable:** $\omega^2 > 0$ and $\omega\xi > 0$
- **Unstable:** $\omega^2 \leq 0$ or $\omega\xi \leq 0$

Can we test $H_0 : \omega^2 \leq 0$ or $\omega\xi \leq 0$ vs $H_1 : \omega^2 > 0$ and $\omega\xi > 0$?

JointODE

└ Motivation

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Damping Effects

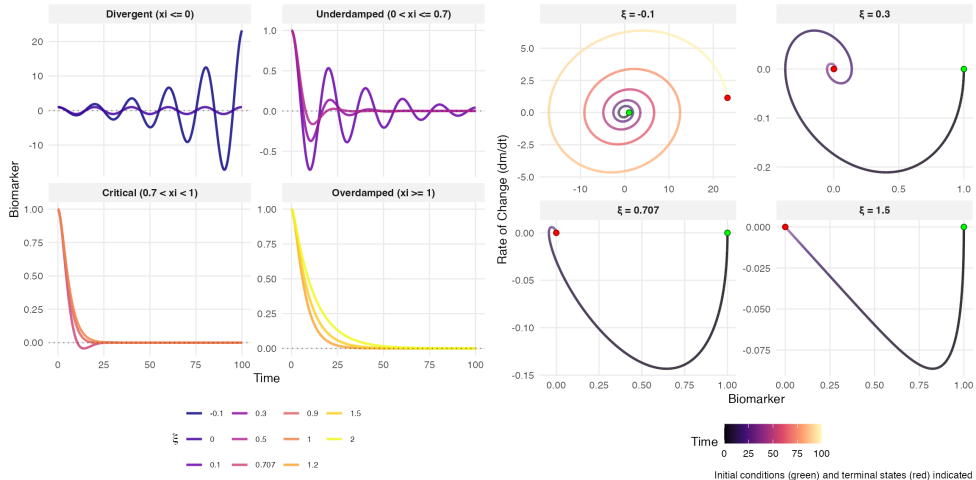
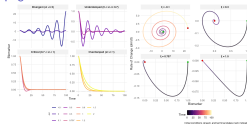


Figure: Autonomous systems with no external excitation ($u(t) = 0$)

Figure: Autonomous systems with no external excitation ($u(t) = 0$)

[Time: 3 min — Total: 10 min]

Let me delve deeper into parameter interpretation. I'll focus on the stable case - where omega-squared is positive - as this provides the clearest insights.

The figure shows the simplest scenario: an autonomous system with constant excitation $u(t)$ equals zero.

Starting with omega - it determines the oscillation period through the relationship T equals two-pi over omega.

In the undamped case where ξ is zero, you can observe complete oscillations roughly every twenty time units.

Now for the damping ratio ξ . The figure illustrates how ξ values create distinct behaviors:

When ξ is negative, the system diverges exponentially with oscillations.

For ξ between zero and zero point seven, we observe decaying oscillations.

Between zero point seven and one point zero, oscillations are minimal - just slight overshooting.

Above one point zero, we enter the overdamped regime - monotonic convergence without any oscillation.

The right panel is the phase plot - biomarker level versus velocity. Notice how the trajectories spiral inward for low ξ , indicating oscillations, and become more direct as ξ increases.

Damping Effects (cont.)

The damping ratio ξ determines convergence behavior

Damping (ξ)	Regime	Behavior
$\xi \leq 0$	Unstable	Divergence or sustained oscillations
$0 < \xi \leq 0.7$	Underdamped	Decaying oscillations
$0.7 < \xi < 1$	Near-critical	Rapid convergence, minimal overshoot
$\xi \geq 1$	Overdamped	Slow monotonic decay

Special cases:

- $\xi = \frac{\sqrt{2}}{2} \approx 0.707$: Fastest settling time with overshoot
- $\xi = 1$: Critical damping, the fastest response without oscillation

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This table summarizes the damping regimes. Each range of ξ produces different dynamics.

Two critical values deserve special attention:

ξ equals root-two over two - provides optimal settling time while allowing some overshoot. This is often desirable in control systems.

ξ equals one represents critical damping - the boundary between oscillatory and non-oscillatory behavior. It gives the fastest response without any oscillation.

External Interventions

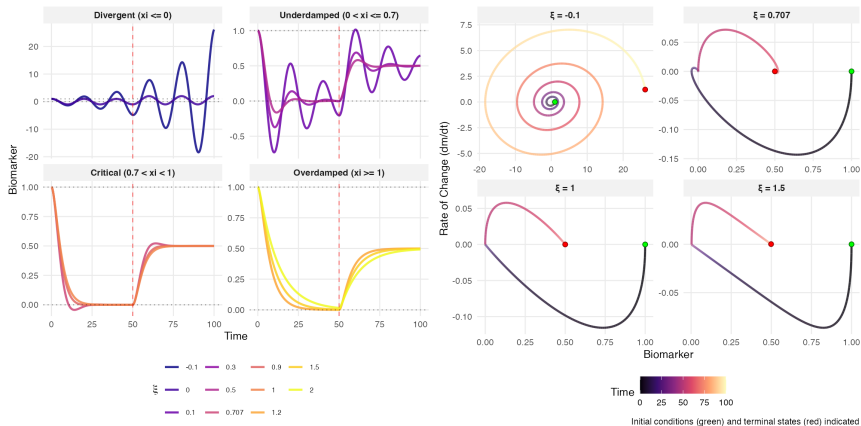


Figure: Non-autonomous systems with step excitation ($u(t) = 0.5 \cdot I\{t \geq 50\}$)

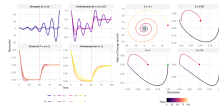


Figure: Non-autonomous systems with step excitation ($u(t) = 0.5 \cdot 1(t \geq 50)$)

[Time: 2 min — Total: 14 min]

Now let's examine the response to external interventions - such as treatment change. Here, at time fifty, we introduce a step change where $u(t)$ jumps from zero to zero point five.

Notice the system's response dynamics. The new equilibrium level equals zero point five.

Importantly, the damping ratio ξ continues to govern the approach to equilibrium - determining whether we see oscillatory or monotonic convergence.

This demonstrates how our ODE framework naturally captures real-world medical interventions - like insulin injections causing glucose levels to drop.

Biological Trajectories Naturally Follow ODE Dynamics

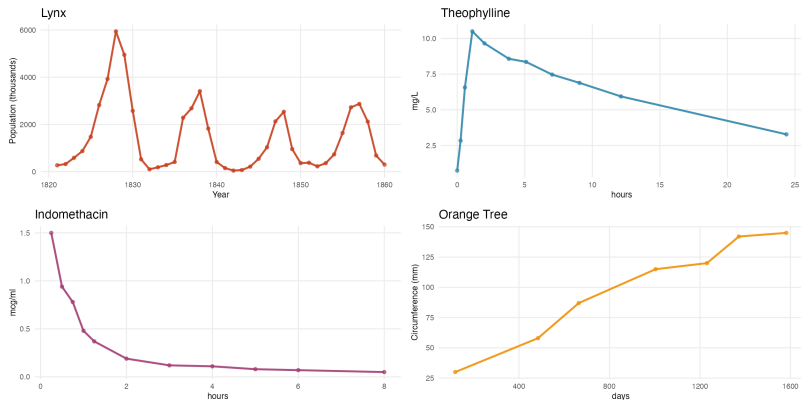


Figure: Diverse biological systems naturally exhibit second-order ODE dynamics. **Top-left:** Lynx population cycles. **Top-right:** Theophylline pharmacokinetics. **Bottom-left:** Indomethacin concentration. **Bottom-right:** Orange tree growth.

2025-09-25

JointODE

└ Motivation

└ Biological Trajectories Naturally Follow ODE Dynamics

[Time: 2 min — Total: 16 min]

Remarkably, second-order ODE dynamics appear throughout biology and medicine. These examples illustrate how widespread such patterns are.

Biological Trajectories Naturally Follow ODE Dynamics

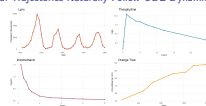


Figure: Diverse biological systems naturally exhibit second-order ODE dynamics. **Top-left:** Lynx population cycles. **Top-right:** Theophylline pharmacokinetics. **Bottom-left:** Indomethacin concentration. **Bottom-right:** Orange tree growth.

Joint Modeling Framework

Longitudinal Submodel:

$$V_{ij} = m_i(T_{ij}) + b_i + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_e^2), \quad b_i \sim \mathcal{N}(0, \sigma_b^2)$$

Biomarker dynamics governed by:

$$\begin{aligned} \ddot{m}_i(t) + 2\xi\omega\dot{m}_i(t) + \omega^2 m_i(t) &= k\omega^2 u_i(t) \Rightarrow \\ \ddot{m}_i(t) &= \beta_1 m_i(t) + \beta_2 \dot{m}_i(t) + \boldsymbol{\beta}_X^\top \mathbf{X}_i(t) \end{aligned}$$

with initial conditions: $m_i(0) = m_{i0}$, $\dot{m}_i(0) = \dot{m}_{i0}$

Survival Submodel:

$$\lambda_i(t) = \lambda_0(t) \exp \left[\underbrace{\alpha_1 m_i(t)}_{\text{level}} + \underbrace{\alpha_2 \dot{m}_i^\gamma(t)}_{\text{volatility}} + \mathbf{W}_i(t)^\top \boldsymbol{\phi} + b_i \right]$$

where $\gamma \in \{1, 2\}$ for linear or squared velocity effects, and $\lambda_0(t)$ is approximated by B-splines $\log(\lambda_0(t)) = \boldsymbol{\eta}^\top \mathbf{B}^{(\lambda)}(t)$

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$$\dot{m}_i(t) = \beta_1 m_i(t) + \beta_2 \dot{m}_i(t) + \beta_3^\top \mathbf{X}_i(t)$$

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[Time: 3 min — Total: 19 min]

With this understanding of ODE dynamics, let me outline our joint modeling framework, which consists of two components.

For the longitudinal submodel, we decompose observations into the true trajectory plus measurement error and random effects. The trajectory evolution follows our second-order ODE, which we reparameterize for computational efficiency.

The survival submodel extends traditional joint models in a key way: we incorporate both the biomarker level through alpha-one and its velocity through alpha-two. The gamma parameter allows flexibility - when gamma equals one, we model linear velocity effects; when gamma equals two, we capture volatility through squared velocity, regardless of direction, which may be more relevant in some clinical contexts, such as heart rate variability.

We approximate the baseline hazard using B-splines to maintain flexibility. The entire framework is estimated via maximum likelihood, though the computational burden exceeds that of standard joint models.

Unified ODE System

Following Tang et al. (2023)³, the survival process can be naturally integrated through the relationship $\frac{d\Lambda(t)}{dt} = \lambda(t)$.

Define the state vector: $\mathbf{s}_i(t) = [\Lambda_i(t), m_i(t), \dot{m}_i(t)]^\top$. The unified system evolves as:

$$\frac{d\mathbf{s}_i}{dt} = \begin{pmatrix} \lambda_i(t|b_i) \\ \dot{m}_i(t) \\ \ddot{m}_i(t) \end{pmatrix}, \quad \text{with} \quad \mathbf{s}_i(0) = \begin{pmatrix} 0 \\ m_{i0} \\ \dot{m}_{i0} \end{pmatrix}$$

Single ODE solver handles both longitudinal and survival processes simultaneously.

³Tang W, et al. J. Am. Stat. Assoc. 2023;118(544):2406-2421.

Following Tang et al. (2023)³, the survival process can be naturally integrated through the relationship $\frac{d\Lambda(t)}{dt} = \lambda(t)$.

Define the state vector: $\mathbf{a}_i(t) = [\Lambda_i(t), m_i(t), \dot{m}_i(t)]^T$. The unified system evolves as:

$$\frac{d\mathbf{a}_i}{dt} = \begin{pmatrix} \lambda_i(t)h_i \\ \dot{m}_i(t) \\ \ddot{m}_i(t) \end{pmatrix}, \quad \text{with } \mathbf{a}_i(0) = \begin{pmatrix} 0 \\ m_{i0} \\ \dot{m}_{i0} \end{pmatrix}$$

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³Tang W, et al. *J. Am. Stat. Assoc.* 2023;118(544):2408-2421.

[Time: 2.5 min — Total: 21.5 min]

Building on Tang and colleagues' two thousand twenty-three framework, we unify both processes into a single ODE system.

The crucial insight is recognizing that the cumulative hazard itself satisfies an ODE: $d\text{-capital-lambda}/dt$ equals little lambda. This allows us to incorporate lambda as an additional state variable.

Consequently, our state vector now comprises three components - the cumulative hazard, biomarker level, and biomarker velocity - all evolving simultaneously. A single ODE solver handles the entire system, which is both elegant and computationally efficient.

Parameter Estimation: EM Algorithm I

We employ the EM algorithm to handle the latent random effects $b_i \sim \mathcal{N}(0, \sigma_b^2)$

Parameters: $\boldsymbol{\theta} = (\omega, \xi, k, \boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\phi}, \boldsymbol{\eta}, \sigma_e^2, \sigma_b^2)$

Observed data: $\mathcal{O}_i = \{(V_{ij}, T_{ij})_{j=1}^{n_i}, T_i, \delta_i\}$

E-step: Given $\boldsymbol{\theta}^{(k)}$, compute posterior moments

$$\hat{b}_i^{(k)} = E[b_i | \mathcal{O}_i; \boldsymbol{\theta}^{(k)}], \quad \hat{v}_i^{(k)} = \text{Var}[b_i | \mathcal{O}_i; \boldsymbol{\theta}^{(k)}]$$

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[Time: 3 min — Total: 27 min]

For parameter estimation, we employ the EM algorithm - the standard approach for joint models with latent random effects.

In the E-step, we compute the posterior mean and variance of the random effects b_i given the observed data and current parameter estimates.

Parameter Estimation: EM Algorithm (cont.)

M-step: Maximize the expected complete-data log-likelihood

$$\begin{aligned} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)}) &= E[\ell_c(\boldsymbol{\theta})|\mathcal{O}; \boldsymbol{\theta}^{(k)}] \\ &= \sum_{i=1}^n E\left[\log f(V_i|b_i; \boldsymbol{\theta}) + \log f(T_i, \delta_i|b_i; \boldsymbol{\theta}) + \log f(b_i; \boldsymbol{\theta}) \middle| \mathcal{O}_i; \boldsymbol{\theta}^{(k)}\right] \end{aligned}$$

Update parameters:

$$\boldsymbol{\theta}^{(k+1)} = \arg \max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(k)})$$

Iterate E and M steps until convergence.

└ Parameter Estimation: EM Algorithm (cont.)

M-step: Maximize the expected complete-data log-likelihood

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$$= \sum_{i=1}^n E \left[\log f(V_i|b_i; \theta) + \log f(T_i, \delta_i|b_i; \theta) + \log f(b_i; \theta) \middle| C_i; \theta^{(k)} \right]$$

Update parameters:

$$\theta^{(k+1)} = \arg \max_{\theta} Q(\theta|\theta^{(k)})$$

Iterate E and M steps until convergence.

[Time: 1 min — Total: 28 min]

In the M-step, we maximize the expected complete-data log-likelihood using the posterior moments from the E-step. This yields updated parameter estimates.

We iterate between E and M steps until convergence.

The primary computational challenge lies in solving ODEs and computing their gradients with respect to parameters. We've developed efficient numerical methods to address this. For those interested in implementation details, I'd be happy to discuss after the presentation.

Simulation Design

Data Generation Process

- **Sample:** $n = 100$ subjects, annual measurements per subject in $[0, 10]$ years
- **Biomarker dynamics:** Second-order ODE with parameters
 - Natural frequency: $\omega = 2\pi/5$ (period = 5 years)
 - Damping ratio: $\xi \in \{0.7, 1.2\}$ (critical/overdamped)
- **Hazard model:** $\lambda_i(t) = \lambda_0(t) \exp[\alpha_1 m_i(t) + \alpha_2 \dot{m}_i(t) + \boldsymbol{\phi}^\top \mathbf{W}_i + b_i]$

Benchmark Comparison

- **Proposed:** Joint ODE model estimating all parameters
- **Oracle:** Cox regression using true $m_i(t_j)$, $\dot{m}_i(t_j)$ at observation times

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[Time: 2 min — Total: 29 min]

Let me present our simulation study, designed as a proof-of-concept for the proposed methodology.

We simulated one hundred subjects over a ten-year period. As a benchmark, we compared our approach against an "oracle" - a Cox model with access to the true biomarker values and velocities at only the observation times.

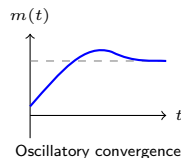
These results are preliminary. We're planning more comprehensive simulation studies once we better characterize real-world data patterns and identify relevant clinical scenarios.

Simulation Results (Critical Damping: $\xi = 0.7$)

Parameter	Method	True	Bias	ESE	ASE	CP(%)
ϕ_1	Oracle	0.40	0.014	0.135	0.127	93
	Proposed		0.018	0.132	0.130	94
ϕ_2	Oracle	-0.60	-0.022	0.261	0.250	94
	Proposed		0.002	0.277	0.252	91
α_1 (level)	Oracle	0.30	0.047	0.259	0.239	92
	Proposed		0.037	0.134	0.132	96
α_2 (velocity)	Oracle	1.00	-0.067	0.643	0.585	92
	Proposed		-0.032	0.218	0.223	96
$T = 2\pi/\omega$	Proposed	5.00	0.008	0.045	0.039	91
ξ	Proposed	0.70	0.000	0.010	0.009	94

Critical Damping

$$\xi = 0.7$$



Note: ESE: empirical SE; ASE: asymptotic SE; CP: coverage probability.

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[Time: 3 min — Total: 32 min]

We first examine the critical damping scenario with ξ equals zero point seven. The true biomarker trajectory converges fast to equilibrium with some overshoot.

The results reveal an interesting pattern. For covariates ϕ_1 and ϕ_2 , our method performs comparably to the oracle - as expected, since these don't involve the ODE structure. But examine the biomarker effects. Our standard errors are smaller than the oracle's. For the level effect α_1 : zero point one three four versus zero point two five nine. For the velocity effect α_2 : zero point two one eight versus zero point six four three.

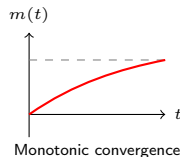
What drives this efficiency gain? The oracle only observes discrete snapshots at observation times. In contrast, our ODE framework models the entire continuous trajectory, effectively interpolating between observations.

Simulation Results (Overdamped: $\xi = 1.2$)

Parameter	Method	True	Bias	ESE	ASE	CP(%)
ϕ_1	Oracle	0.40	0.011	0.136	0.131	93
	Proposed		0.015	0.136	0.134	94
ϕ_2	Oracle	-0.60	-0.033	0.276	0.258	92
	Proposed		-0.023	0.280	0.260	92
α_1 (level)	Oracle	0.30	0.040	0.258	0.238	92
	Proposed		0.042	0.159	0.151	94
α_2 (velocity)	Oracle	1.00	-0.105	1.070	0.943	90
	Proposed		-0.138	0.449	0.412	94
$T = 2\pi/\omega$	Proposed	5.00	0.015	0.086	0.083	92
ξ	Proposed	1.20	-0.003	0.026	0.027	94

Overdamped

$\xi = 1.2$



Note: ESE: empirical SE; ASE: asymptotic SE; CP: coverage probability.

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	Proposed		0.015	0.136	0.134	94
ϕ_2	Oracle	-0.60	-0.033	0.276	0.258	92
	Proposed		-0.023	0.280	0.260	92
α_1 (level)	Oracle	0.30	0.040	0.258	0.238	92
	Proposed		0.042	0.159	0.151	94
α_2 (velocity)	Oracle	1.00	-0.105	1.070	0.943	90
	Proposed		-0.138	0.449	0.412	94
$T = 2\pi/\omega$	Proposed	5.00	0.015	0.086	0.083	92
	Proposed	1.20	-0.003	0.026	0.027	94

Note: ESE: empirical SE; ASE: asymptotic SE; CP: coverage probability.



[Time: 2 min — Total: 34 min]

Here are the results for the overdamped settings, where ξ equals one point two. The biomarker converges slowly and monotonically to equilibrium.

The overdamped scenario exhibits similar efficiency gains for biomarker effects.

However, note that velocity estimation becomes more challenging - the standard errors for α_2 are larger than in the critical damping case. This is intuitive: slowly evolving systems provide less information about rates of change.

Extrapolation Capability of Proposed Method

man/figures/README-visualization-1.png

[Time: 2 min — Total: 36 min]

A notable advantage of the ODE framework is its natural extrapolation capability. Since we model the mechanistic dynamics, we can predict biomarker levels and velocities even beyond the observed time window.

This figure demonstrates individual trajectory predictions from our model. Observe how the framework provides extrapolation beyond the observation window.

Summary

Methodological Innovation:

1. Mechanistic modeling of biomarker trajectories via second-order ODEs
2. Natural incorporation of velocity

Future Directions:

- **Heterogeneous dynamics:** Allow patient-specific ω and ξ parameters to capture population heterogeneity
- **Variable selection:** Develop methods for selecting informative biomarkers when multiple candidates are available
- **Biomarker interactions:** Model cross-talk and feedback loops between multiple biomarkers via coupled ODEs

1. Mechanistic modeling of biomarker trajectories via second-order ODEs
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[Time: 2.5 min — Total: 38.5 min]

Let me summarize our current work and future directions.

We've developed a mechanistic joint modeling framework using second-order ODEs that offers several advantages: - natural incorporation of biomarker velocity - interpretable parameters with physical meaning

Our future research will focus on three areas: allowing patient-specific dynamic parameters, extending to multiple correlated biomarkers, and identifying specific clinical contexts where velocity information provides the greatest predictive value.

Computational Challenges

Individual ODE Solutions

- Each patient requires solving their own ODE system
- Computational cost scales linearly with sample size: $\mathcal{O}(n)$

Gradient Computation via Augmented ODEs

- Computing $\nabla_{\theta} \mathcal{L}$ requires solving an augmented system:

$$\frac{d}{dt} \begin{pmatrix} \mathbf{s}(t) \\ \frac{\partial \mathbf{s}}{\partial \boldsymbol{\theta}} \end{pmatrix} = \begin{pmatrix} f(\mathbf{s}, \boldsymbol{\theta}, t) \\ \frac{\partial f}{\partial \mathbf{s}} \frac{\partial \mathbf{s}}{\partial \boldsymbol{\theta}} + \frac{\partial f}{\partial \boldsymbol{\theta}} \end{pmatrix}$$

- Dimension grows from d to $d(1 + p)$ where p = number of parameters
- Significant memory and computation overhead

JointODE

└ Conclusions

└ Computational Challenges

Computational Challenges

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- Computational cost scales linearly with sample size: $O(n)$

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- Dimension grows from d to $d(1 + p)$ where p = number of parameters
- Significant memory and computation overhead

[Time: 2.5 min — Total: 41 min]

I should address the computational challenges inherent in this approach.

First, scalability: each patient requires solving their own ODE system, leading to $O(N)$ computational complexity. For large cohort studies, this can become prohibitive.

Second, gradient computation: to optimize the likelihood, we must compute nabla (pronounced "NAB-lah") theta L, which requires solving augmented ODE systems that track both states and their parameter sensitivities.

This augmentation increases the system dimension from D to D times (one plus P), where P is the number of parameters. For complex models, this creates substantial computational overhead.

We're actively investigating several solutions: parallelization across patients, adaptive ODE solvers that adjust precision dynamically, and automatic differentiation frameworks that may provide more efficient gradient computation.

Thank You

Questions & Discussion

Resources:

`github.com/ziyangg98/JointODE`

Thank you for your attention and feedback

[Time: 2 min — Total: 43 min + QA]

Thank you for your attention.

Our code is publicly available on GitHub at [GitHub dot com slash ziyangg ninety-eight slash joint ODE](https://github.com/ziyangg98/JointODE). We're currently preparing a manuscript with full technical details.

I welcome any questions about the methodology, computational aspects, or potential applications to your own research problems.

[Be prepared for questions on: - How to initialize the EM algorithm - Handling missing data patterns - Extension to multiple biomarkers - Computational efficiency strategies - Software implementation details - Performance versus standard joint models]

Appendix: ODE Solution Forms and Damping Behaviors

Damping	Condition	Solution Form	Behavior
Underdamped	$0 < \xi < 1$	$e^{-\xi\omega t}[A \cos(\omega_d t) + B \sin(\omega_d t)]$	Decaying oscillations
Critical	$\xi = 1$	$(A + Bt)e^{-\omega t}$	Fastest decay
Overdamped	$\xi > 1$	$Ae^{\lambda_1 t} + Be^{\lambda_2 t}$	Slow decay
Undamped	$\xi = 0$	$A \cos(\omega t) + B \sin(\omega t)$	Sustained oscillations
Unstable	$\omega^2 < 0$ or $\omega\xi < 0$	Exponential growth	System diverges

- $\omega_d = \omega\sqrt{1 - \xi^2}$: damped natural frequency
- $\lambda_{1,2} = -\xi\omega \pm \omega\sqrt{\xi^2 - 1}$: overdamped eigenvalues
- A, B : constants determined by initial conditions $m_0, \dot{m}_0$

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- $\lambda_{1,2} = -\xi\omega \pm \omega\sqrt{\xi^2-1}$: overdamped eigenvalues
- A, B : constants determined by initial conditions $m_0, \dot{v}_0$

[Time: 2 min if asked]

[If asked about mathematical details]

This table shows the analytical solution forms for different damping regimes.

For underdamped systems where ξ is between zero and one, we get e to the negative $\xi\omega t$ [read: e to the negative ξ omega t] times oscillating terms with cosine and sine of $\omega_d t$ [read: omega d t].

Critical damping at ξ equals one gives the fastest decay without oscillation.

Overdamped systems have two real eigenvalues λ_1 and λ_2 [read: λ_1 one and λ_2 two].

The damped natural frequency ω_d equals ω times square root of one minus ξ^2 [read: omega d equals omega times square root of one minus ξ squared].

Appendix: Model Reparameterization

We reparameterize the ODE for easier estimation:

$$\ddot{m}(t) + 2\xi\omega\dot{m}(t) + \omega^2m(t) = k\omega^2u(t)$$

↓ Reparameterization

$$\ddot{m}(t) = \beta_1m(t) + \beta_2\dot{m}(t) + \boldsymbol{\beta}_X^\top\mathbf{X}(t)$$

Parameter Recovery:

$$\omega = \sqrt{-\beta_1}$$

$$\xi = -\beta_2/(2\sqrt{-\beta_1})$$

$$\text{SE via Delta method: } \nabla g(\boldsymbol{\beta})^\top \text{Var}(\hat{\boldsymbol{\beta}}) \nabla g(\boldsymbol{\beta})$$

Appendix: Model Reparameterization

[Time: 2 min if asked]

[If asked about parameter estimation details]

We reparameterize the ODE for easier estimation. The original form has products like two-xi-omega [read: two xi omega] which are hard to optimize.

After reparameterization, beta-one equals negative omega-squared [read: beta one equals negative omega squared], and beta-two equals negative two- xi-omega [read: beta two equals negative two xi omega].

To recover the original parameters: omega equals square root of negative beta-one [read: omega equals square root of negative beta one], and xi equals negative beta-two over two times square root of negative beta-one [read: xi equals negative beta two over two times square root of negative beta one].

Standard errors are computed via the delta method using the gradient of the transformation.

$$\ddot{u}(t) + 2\xi\omega\dot{u}(t) + \omega^2u(t) = k\omega^2u(t)$$

↳ Reparameterization

$$\ddot{u}(t) = \beta_1\dot{u}(t) + \beta_2u(t) + \beta_3\mathbf{X}(t)$$

Parameter Recovery:

$$\omega = \sqrt{-\beta_1}$$

$$\xi = -\beta_2/(2\sqrt{-\beta_1})$$

$$\text{SE via Delta method: } \nabla g(\beta)^T \text{Var}(\hat{\beta}) \nabla g(\beta)$$

Appendix: EM Algorithm

1. **Initialization:** Set $\theta^{(0)}$, $k \leftarrow 0$
2. **Main Loop:**
 - 2a. *ODE phase:* For each patient i
 - Solve augmented ODE system $\rightarrow (m_i, \frac{\partial m_i}{\partial \theta})$
 - Compute hazard components
 - 2b. *Posterior phase:* Compute $\hat{b}_i$, $E[e^{b_i}]$
 - 2c. *Gradient phase:* Evaluate $\nabla_{\theta} Q$
 - 2d. *Update:* $\theta^{(k+1)} \leftarrow$ L-BFGS-B step
3. **Convergence:** $\|\nabla Q\| < 10^{-4}$ or $|\Delta Q| < 10^{-6}$

Appendix: EM Algorithm

[Time: 2-3 min if asked]

[If asked about algorithm details]

Here's our EM algorithm structure.

We initialize parameters θ_0 [read: theta naught] using method of moments or simple mixed models.

In each iteration: First, the ODE phase solves the augmented system for each patient to get trajectories (tra-JECK-tor-eez) and sensitivities. Second, the posterior phase computes $\hat{b}_i$ [read: b hat sub i] and the expectation of e to the b_i [read: e to the b sub i]. Third, we evaluate gradients of the Q function. Fourth, we update using L-BFGS-B, a limited-memory quasi-Newton method.

We converge when the gradient norm is less than ten to the minus four [read: ten to the minus four] or the change in Q is less than ten to the minus six [read: ten to the minus six].

1. **Initialization:** Set $\theta^{(0)}$, $k \leftarrow 0$
2. **Main Loop:**
 - 2a. **ODE phase:** For each patient i
 - Solve augmented ODE system $\rightarrow (m_i, \frac{dy_i}{dt})$
 - Compute hazard components
 - 2b. **Posterior phase:** Compute $\hat{b}_i, E[e_i^{b_i}]$
 - 2c. **Gradient phase:** Evaluate $\nabla_{\theta} Q$
 - 2d. **Update:** $\theta^{(k+1)} \leftarrow \text{L-BFGS-B step}$
3. **Convergence:** $\|\nabla Q\| < 10^{-4}$ or $|\Delta Q| < 10^{-6}$

Appendix: Posterior Computation

Key Insight - Factorization:

$$\lambda_i(t|b_i) = e^{b_i} \lambda_i(t|0), \quad \Lambda_i(t|b_i) = e^{b_i} \Lambda_i(t|0)$$

Computational Strategy:

1. Solve ODE once with $b_i = 0$ to get $\Lambda_i(t|0)$
2. Find posterior mode by maximizing:

$$\ell_i(b) = b \left[\frac{S_i}{\sigma_e^2} + \delta_i \right] - \frac{b^2}{2} \left[\frac{n_i}{\sigma_e^2} + \frac{1}{\sigma_b^2} \right] - e^b \Lambda_i(T_i|0)$$

where $S_i = \sum_j (V_{ij} - m_i(T_{ij}))$

3. Compute moments via Gauss-Hermite quadrature

└ Appendix: Posterior Computation

[Time: 2 min if asked]

[If asked about computational efficiency]

The key insight is factorization - the hazard with random effect b_i equals e to the b_i [read: e to the b sub i] times the hazard at b_i equals zero.

This allows us to: First, solve the ODE once with b_i equals zero. Second, find the posterior mode by maximizing this log-posterior, where s_i [read: S sub i] is the sum of residuals. Third, compute moments using Gauss-Hermite quadrature around the mode. This factorization dramatically reduces computation - we don't need to re-solve ODEs for different b_i values.

$$\lambda_i(t|b_i) = e^{b_i} \lambda_i(t|0), \quad A_i(t|b_i) = e^{b_i} A_i(t|0)$$

1. Solve ODE once with $b_i = 0$ to get $A_i(t|0)$
2. Find posterior mode by maximizing:

$$\ell_i(b) = b \left[\frac{S_i}{\sigma_i^2} + \hat{b}_i \right] - \frac{b^2}{2} \left[\frac{b_i}{\sigma_i^2} + \frac{1}{\sigma_i^2} \right] = e^{b_i} A_i(T_i|0)$$

where $S_i = \sum_j (V_{ij} - m_i(T_{ij}))$

3. Compute moments via Gauss-Hermite quadrature

Appendix: Gradient-Based Optimization

Objective: Maximize $Q(\boldsymbol{\theta})$ where $\boldsymbol{\theta} = (\boldsymbol{\eta}, \boldsymbol{\phi}, \boldsymbol{\alpha}, \boldsymbol{\beta}, \sigma_e^2, \sigma_b^2)$

Gradient Decomposition:

$$\nabla_{\boldsymbol{\theta}} Q = \underbrace{\sum_{i,j} \frac{r_{ij}}{\sigma_e^2} \frac{\partial m_i(T_{ij})}{\partial \boldsymbol{\theta}}}_{\text{Longitudinal component}} + \underbrace{\sum_i \left[\delta_i \frac{\partial \log \lambda_i}{\partial \boldsymbol{\theta}} - E[e^{b_i}] \frac{\partial \Lambda_i}{\partial \boldsymbol{\theta}} \right]}_{\text{Survival component}}$$

where $r_{ij} = V_{ij} - m_i(T_{ij}) - \hat{b}_i$ is the residual

No analytical form $\rightarrow$ numerical computation required

Appendix: Gradient-Based Optimization

[Time: 2 min if asked]

[If asked about optimization details]

The gradient of Q decomposes into two parts.

The longitudinal component involves residuals r_{i-j} [read: r sub i j] weighted by one over σ_e^2 , times the trajectory sensitivities.

The survival component has the event indicator δ_i times the log-hazard gradient, minus the expected cumulative hazard gradient.

Since we don't have analytical forms for the ODE solutions, all gradients must be computed numerically by solving the augmented ODE system.

$$\nabla_{\theta} Q = \underbrace{\sum_{i,j} \frac{r_{ij}}{\sigma_e^2} \frac{\partial m_i(T_{ij})}{\partial \theta}}_{\text{Longitudinal component}} + \underbrace{\sum_i \left[\delta_i \frac{\partial \log \lambda_i}{\partial \theta} - E[\delta_i] \frac{\partial \lambda_i}{\partial \theta} \right]}_{\text{Survival component}}$$

where $r_{ij} = V_{ij} - m_i(T_{ij}) = \hat{b}_i$ is the residual

No analytical form $\rightarrow$ numerical computation required

Appendix: Gradient-Based Computation cont.

Strategy: Solve state and sensitivities Simultaneously

Denote $\mathbf{Z}_i(t) = [m_i(t), \dot{m}_i(t), \mathbf{X}_i(t)^\top]^\top$, then the augmented system is:

$$\frac{d}{dt} \begin{pmatrix} \Lambda_i(t) \\ m_i(t) \\ \dot{m}_i(t) \\ \frac{\partial \Lambda_i}{\partial \boldsymbol{\beta}} \\ \frac{\partial m_i}{\partial \boldsymbol{\beta}} \\ \frac{\partial \dot{m}_i}{\partial \boldsymbol{\beta}} \end{pmatrix} = \begin{pmatrix} \lambda_i(t|0) \\ \dot{m}_i(t) \\ \boldsymbol{\beta}^\top \mathbf{Z}_i(t) \\ \frac{\partial \lambda_i}{\partial \boldsymbol{\beta}} + \lambda_i \boldsymbol{\alpha}^\top \frac{\partial \mathbf{m}_i}{\partial \boldsymbol{\beta}} \\ \frac{\partial \dot{m}_i}{\partial \boldsymbol{\beta}} \\ \mathbf{Z}_i(t) + \boldsymbol{\beta}^\top \frac{\partial \mathbf{Z}_i(t)}{\partial \boldsymbol{\beta}} \end{pmatrix}$$

Initial State $(0, m_{i0}, \dot{m}_{i0})$, Sensitivities all zero

└ Appendix: Gradient-Based Computation cont.

$$\frac{d}{dt} \begin{pmatrix} \Lambda_i(t) \\ m_i(t) \\ \dot{m}_i(t) \\ \frac{\partial \Lambda_i}{\partial \alpha} \\ \frac{\partial \dot{m}_i}{\partial \alpha} \end{pmatrix} = \begin{pmatrix} \lambda_i(t|0) \\ \dot{m}_i(t) \\ \beta^T \mathbf{Z}_i(t) \\ \frac{\partial \Lambda_i}{\partial \alpha} + \lambda_i \alpha^T \frac{\partial \Lambda_i}{\partial \alpha} \\ \mathbf{Z}_i(t) + \beta^T \frac{\partial \mathbf{Z}_i}{\partial \alpha} \end{pmatrix}$$

[Time: 2-3 min if asked]

[If asked about implementation details]

We solve states and sensitivities simultaneously in one augmented ODE system.

The state vector $\mathbf{z}_i$ [read: Z sub i] contains the biomarker level, velocity, and covariates.

The augmented system has the original three states plus their sensitivities with respect to all parameters.

The initial state starts at zero for cumulative hazard, m_{i0} and $\dot{m}_{i0}$ [read: m sub i naught and m dot sub i naught] for the biomarker.

All sensitivities start at zero since initial conditions don't depend on parameters.

This approach is more efficient than finite differences and provides exact gradients up to ODE solver tolerance.